RingMo-Qlite: A Fully Quantized Lightweight Network for Efficient Remote Sensing Image Interpretation

Yingxuan Long, Liangjin Zhao, Tong Liu, Ziyang Yuan, Wenhui Diao, Menglong Yan, and Zhirui Wang

***Abstract*—With the development of remote sensing satellite technology, on-orbit interpretation of remote sensing images has become a new trend. Hybrid network architectures, such as those combining convolutional neural networks (CNNs) and transformers, have shown great potential in remote sensing image interpretation due to their ability to capture local features and long-range dependencies. However, these architectures face limitations in real-time on-orbit processing due to their large parameter sizes and high computational complexity. To address this, we propose RingMo-Qlite, a fully quantized remote sensing model designed for efficient processing. All weights, activations and nonlinear layers are quantized to low-bit integers, with inference relying solely on integer and shift operations. We introduce a training-free token mixer to replace the multihead self-attention mechanism in the transformer branch, effectively reducing precision loss during quantization. For the CNN branch's high-frequency feature extraction, we employ a reparameterization strategy that simplifies the multibranch structure used during training to a single-branch structure during inference, significantly reducing computational load. In addition, we design a novel quantization strategy that includes a peak-aware quantizer and an outlier-robust observer, tailored to the complex distribution of remote sensing images to minimize precision loss. Experiments on various remote sensing scene classification datasets demonstrate that RingMo-Qlite achieves higher accuracy than existing lightweight methods, with near-lossless performance under INT8 quantization (less than 1% accuracy loss) and comparable performance under INT4 quantization.**

***Index Terms*—Convolutional neural network (CNN)-transformer hybrid framework, model quantization, posttraining quantization (PTQ), remote sensing frequency domain features, reparameterization.**

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I. Introduction

In RECENT years, remote sensing images have been widely used in urban planning [1], [2], environmental monitoring [3], [4], [5], natural disaster detection [6], [7], [8], and geospatial object detection [9], [10], [11]. However, the real-time data processing capabilities of remote sensing models deployed on satellites are limited by on-board hardware resources. Meanwhile, the exponential growth of remote sensing data, reaching petabytes per day, has far exceeded the processing capacity of traditional models [12], [13]. Therefore, developing efficient and accurate real-time processing methods for on-orbit remote sensing data has become a key research direction in remote sensing.

Although some efforts have been made to design lightweight remote sensing models [14], [15], these models still rely heavily on floating-point operations (FLOPs) and consume significant computing resources. In contrast, INT8 quantization can significantly reduce model size and computational complexity. Compared to FP32 operations, INT8 operations can save up to 30 times the energy consumption and 116 times the chip area [16], providing a critical path for model deployment in resource-constrained scenarios.

However, applying quantization to existing remote sensing models like [17] faces multiple challenges, which are further exacerbated by the combination of architecture characteristics and remote sensing data properties.

First, the self-attention mechanism, a core component of the transformer, is inherently difficult to quantize and carries significant computational overhead. From a quantization perspective, self-attention involves multiple nonlinear transformations, such as Softmax and LayerNorm (LN), which are extremely sensitive to numerical precision. Small differences in floating-point values are amplified in these nonlinear processes, and the discretization introduced by quantization directly disrupts this delicate numerical mapping. This leads most existing schemes to retain FLOPs for these nonlinear layers, requiring redundant dequantization-quantization round-trip and additional FP32 buffers as illustrated in Fig. 1(a). From a computational perspective, multihead self-attention itself is computationally intensive. Even fully quantized, it still imposes a heavy computational and memory burden on high-resolution remote sensing feature maps, severely restricting the deployment of quantized models in resource-constrained environments.

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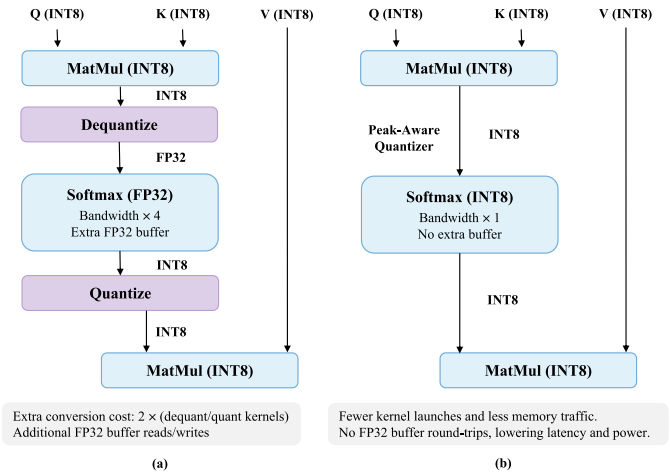


Fig. 1. Comparison of different quantization schemes for the attention mechanism in transformer architecture. (a) Simulates quantization scheme: all operations use floating-point arithmetic. Weights stored as INT, but activations are dequantized to FP32 for Softmax and requantized, incurring two extra conversions and FP32 buffers. (b) Proposed full-integer quantization scheme: End-to-end INT8 without float round-trips; Softmax uses a peak-aware quantizer for stable INT8 calibration.

Second, when the model is fully quantized to low bits, the convolutional neural network (CNN) module's ability to capture high-frequency details is significantly reduced. Low-bit discretization significantly compresses the numerical representation space, making it impossible for CNN to accurately depict high-frequency information in remote sensing images, such as ground texture, edge gradients, and small target outlines, and such information is the key to distinguishing farmland from buildings, roads from water bodies.

Furthermore, directly applying the quantization strategies of existing solutions to remote sensing scenarios presents challenges. Remote sensing images often exhibit outliers due to sensor noise, cloud reflections, and specific ground features, significantly distorting data statistics, and then leading to significant deviations in global quantization parameters. Existing schemes (such as FQ-ViT [18]) typically use the L2 norm to calculate the error between the original and quantized values, which is highly sensitive to outliers, forcing parameter calibration to overfit these outlier samples. Second, inspired by Bondarenko et al. [19], the presoftmax layer (critical for self-attention) has an expanded range to refine attention scores. By observing the numerical distribution of remote sensing scenes, as shown in Fig. 2, we find that the presoftmax layer has a large numerical range, making it difficult to quantize and containing high-value peaks that carry key semantic information. However, existing quantization schemes use a single global (scale and zero-point) pair, whose quantization interval is dominated by the densely distributed main area, leading to detail loss of the already sparse high-value peaks due to the large quantization interval.

To address these challenges, we propose RingMo-Qlite. Our RingMo-Qlite is a fully quantized remote sensing network that quantizes all modules and performs inference using only integer and shift operations. As shown in Fig. 1, unlike other quantization methods, we quantize nonlinear layers like the attention mechanism. Building on the CNN-transformer dual-branch structure, we introduce a new high-/low-frequency information extraction module designed to achieve fast inference while maintaining performance. In addition, we develop a low-bit quantization method tailored to the numerical distribution of remote sensing data.

Specifically, to reduce the computational burden of the self-attention mechanism and prevent precision loss during quantization, we replace the self-attention mechanism with a parameter-free average pooling layer in the first two stages, which aggregates feature information by averaging all elements within each window, effectively extracting low-frequency features from remote sensing images while simplifying model complexity and avoiding precision degradation during quantization. Furthermore, to efficiently capture high-frequency details in a fully quantized network, we adopt a reparameterization strategy that decouples training and inference. Inspired by Zhao et al. [20], during training, a multi-branch architecture, including identity mapping, medium-/small-core convolution branches, and a parameter-free max pooling branch, is used to perceive high-frequency details. These branches are optimized into a streamlined structure during inference through reparameterization, achieving a balance between speed and accuracy.

To address the challenge inherent to the transformer pre-softmax layer, we introduce a peak-aware quantizer, which identifies and partitions data into base and peak regions via targeted constraints. We prioritize allocating quantization resources to the peak region and use independent observers to calibrate region-specific parameters. This approach not only mitigates the issue of excessively coarse quantization intervals for high-value peaks, as such intervals are an inherent limitation of global quantization frameworks dominated by base data distributions, but also preserves the key semantic information in these peaks, thus overcoming the shortcomings of traditional uniform quantization schemes.

To mitigate the impact of remote sensing image outliers on quantization accuracy, we propose the outlier-robust observer, adopting an adaptive parameter optimization method. It addresses the L2 norm's outlier sensitivity by integrating Huber loss with iterative adjustment. Huber loss balances normal data precision and outlier robustness using L2-like squared error for small discrepancies and L1-like linear error for large deviations. Combined with iterative refinement, this reduces outlier-induced global parameter biases and ensures numerical stability in full quantization.

The article's contributions can be summarized as follows.

1) We propose a training-free token mixer that effectively extracts low-frequency features, immune to precision loss during full quantization. This design ensures robust feature extraction and maintains high computational efficiency.
2) We introduce a high-frequency feature extraction module that employs a reparameterization strategy to simplify the multibranch structure used during training into a single-branch structure at inference time. This design ensures efficient and accurate detail extraction, even after full quantization.

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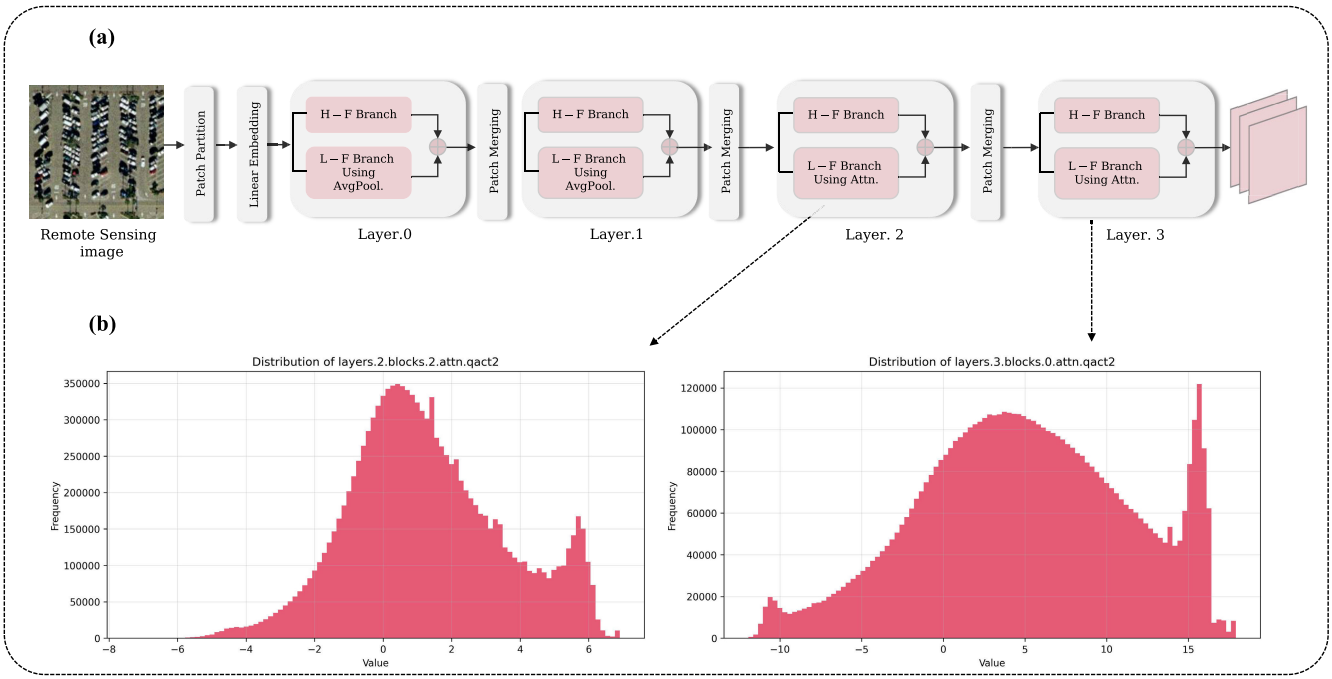


Fig. 2. Numerical distribution histogram of transformer presoftmax layer. (a) Schematic of remote sensing image processing flow, where H–F branch refers to HFB; L–F branch refers to the LFB. (b) Numerical distribution of presoftmax layer, displaying the statistical distribution of numerical values in the presoftmax layer.

3) We design a dedicated low-bit quantization scheme designed for the numerical distribution of remote sensing scenarios, which mitigates the loss of peak features in the transformer presoftmax layer via a peak-aware quantizer, resists outlier interference through an outlier-robust observer, and significantly enhances model robustness and efficiency. The loss in precision is less than 1.5% under the quantization of INT8, and satisfactory performance is achieved under the quantization of INT4.

II. Related Work

A. Lightweight Model Design

In the field of deep learning, the development of lightweight foundational models has emerged as a crucial research focus, especially in resource-limited settings where computational capacity and energy consumption are severely restricted. Given the remarkable success of deep learning models in diverse image processing tasks, adapting these sophisticated models to edge devices poses a considerable challenge.

Vision transformer (ViT) [21] adapts transformers, which are successful in natural language processing, to images by converting them into fixed patches processed via self-attention. Swin transformer [22] increases efficiency with hierarchical window-localized self-attention. Yet both, relying on self-attention, have massive parameters, hindering edge deployment due to high computation demands.

Previous lightweight design approaches have included reductions in model size and adjustments to the number of layers, integration of efficient convolutional techniques such as depthwise separable convolutions, and optimizations of self-attention structures. MobileViT [23], designs for mobile applications, lowers computational complexity by reducing model depth, width, and the number of attention heads. TinyViT [24] minimizes model size by streamlining the number of layers and the dimensions of the hidden layers. EfficientViT [25] employs depthwise separable convolutions to reduce the number of parameters and computational load, ensuring the preservation of effective feature representation. While these approaches provide a certain degree of parameter reduction, they still maintain the multihead self-attention structure, which remains a significant computational challenge, especially in resource-constrained environments. Recent studies have attempted to refine the multihead self-attention mechanism. For example, Pan et al. [26] introduce the use of multilayer perceptrons (MLPs) for initial feature extraction. Wang et al. [27] reduce the spatial dimensions of the attention keys and the value matrices. However, these modifications still require a significant number of parameters, which can lead to performance loss after low-bit quantization. PoolFormer [28] challenges the necessity of the token mixer module, such as self-attention, in the transformer architecture by replacing it with a simple nonparameter spatial average pooling layer. This replacement can significantly reduce computational costs without affecting quantization performance. However, this approach may not be sufficient to capture global dependencies when applied to all stages of feature extraction in remote sensing scenarios. To address this, we adopt the philosophy of PoolFormer [28] by incorporating average pooling layers as token mixers in the early stages of

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feature extraction within transformer branches, while retaining deep self-attention structures to capture complex semantics and long-range dependencies, which are crucial for remote sensing applications.

Unlike the requirements of natural scenes, to ensure performance, lightweight networks in remote sensing scenes must be able to efficiently extract local and global features. LTNet [14] proposes a lightweight transformer network, utilizing a multilevel group convolution (MLGC) module to capture global dependencies with low computational costs. It boosts local feature diversity and improves classification accuracy. Targeting the limited interwindow connections and heavy computation of ViT, LDBST [15] designs a dual-branch structure that combines ViT and CNN branches, building on the hierarchical Swin transformer. Recent work [29] incorporates a transformer-based model, which is tailored for real-time inference on UAV platforms through the application of efficient posttraining quantization (PTQ) techniques. Jiang et al. [30] propose an efficient PTQ method for CNN-based object detectors. However, the aforementioned methods do not consider distinguishing between high and low frequency features in remote sensing scenarios, and the designed modules are highly sensitive to quantization.

Inspired by some works [20], [31] that utilize reparameterization techniques for rapid network inference, our work aims to efficiently capture both high- and low-frequency image information while employing reparameterization techniques to design a more lightweight network inference process, maintaining high performance in various remote sensing image interpretation downstream tasks even under full quantization.

B. Model Quantization

In the field of model quantization, previous studies have focused on two main strategies: PTQ and quantization-aware training (QAT). To reduce the high cost of quantization and avoid retraining with the entire dataset, PTQ has gained greater attention, leading to a series of quantization efforts targeting different architectures. Within the remote sensing domain, CNNs dominated early deep learning models. This is due to their mature deployment pipelines and efficiency in local feature extraction. Their quantization research emerged earlier and saw broader application. These CNN-based quantization methods are highly compatible with classic remote sensing tasks.

Among remote sensing CNN quantization studies, GradQuant [32] mitigates quantization accuracy degradation caused by both the clipping of outlier activations and insufficient overall feature mining. Wei et al. [33] present a mixed-precision quantization method including a symmetric nonlinear scheme and a three-step training strategy. Zhang et al. [34] put forward a hybrid quantization module and a one-to-one self-teaching module to reduce quantization loss. Li et al. [35] propose symmetrically progressive decreasing mixed-precision quantization method to improve the performance of infrared small target segmentation. However, these methods only adopt the CNN architecture, which cannot effectively extract long-range dependencies in remote sensing images, and most of them apply mixed-precision quantization, failing to complete quantization using only limited bit-widths.

With the rise of ViTs, their ability to capture long-range dependencies has offered new solutions for complex remote sensing scenarios. Unlike CNN quantization, the core challenges of ViT quantization lie in addressing unique numerical distributions. Q-ViT [36] proposes a fully differentiable quantization method for ViT, treating quantization bit-width and scales as learnable parameters and introducing a switchable scale technique, pushing the quantization limit of ViT to 3 bits. PTQ4ViT [37] proposes a twin uniform quantization method and uses Hessian-guided metrics to evaluate different scaling factors, achieving near-lossless prediction accuracy on the ImageNet classification task with 8-bit quantization. FQ-ViT [18] finds that the interchannel variation in LN inputs is severe and the attention map values have an extremely uneven distribution, introducing quadratic scale quantization and log-int quantization methods for LN and Softmax, respectively. RepQ-ViT [38] decouples the quantization and inference processes, with the former employing complex quantizers and the latter using scale reparameterization simplified quantizers, focusing on addressing the extreme distribution of LN and Softmax activations and enhancing the accuracy of 4-bit PTQ of ViT to a usable level. I&S-ViT [39] proposes a shift-uniform-log2 quantizer (SULQ) to accurately approximate the distribution of postsoftmax activations and a three-stage smooth optimization strategy to further improve the accurate quantization of weights and activations in the network. I-ViT [40] utilizes integer operations to implement nonlinear operations in ViT, proposing integer shift operations for Softmax and GELU, enabling quantized models to fully leverage fast and efficient low-precision integer arithmetic units.

Notably, these general models are not suitable for remote sensing due to inherent limitations. They are sensitive to remote sensing-specific outliers via L2 norm-based calculation or global quantization, fail to preserve sparse high-value peaks in remote sensing transformers' presoftmax layer with a single (scale and zero-point) pair, and lack optimization for hybrid architecture CNN branches, degrading high-frequency feature extraction. I-ViT [40], though supporting integer inference, is designed for natural scenarios, unable to capture dense high-frequency features, and relies on retraining-required QAT. Most of the general models also use dequantized floating-point computations, underutilizing low-precision units.

Our work integrates CNN and transformer strengths, drawing on integer-only inference while adopting PTQ to avoid retraining. We propose a fully quantized network, enabling model inference via integer and shift operations. Targeting remote sensing data characteristics, our method addresses the above gaps with dedicated designs for enhanced practicality and real-time performance.

III. Methodology

A. Overview

The architectural framework of our proposed model is depicted in Fig. 3. As detailed in the network configuration presented in Table I, our RingMo-Qlite architecture

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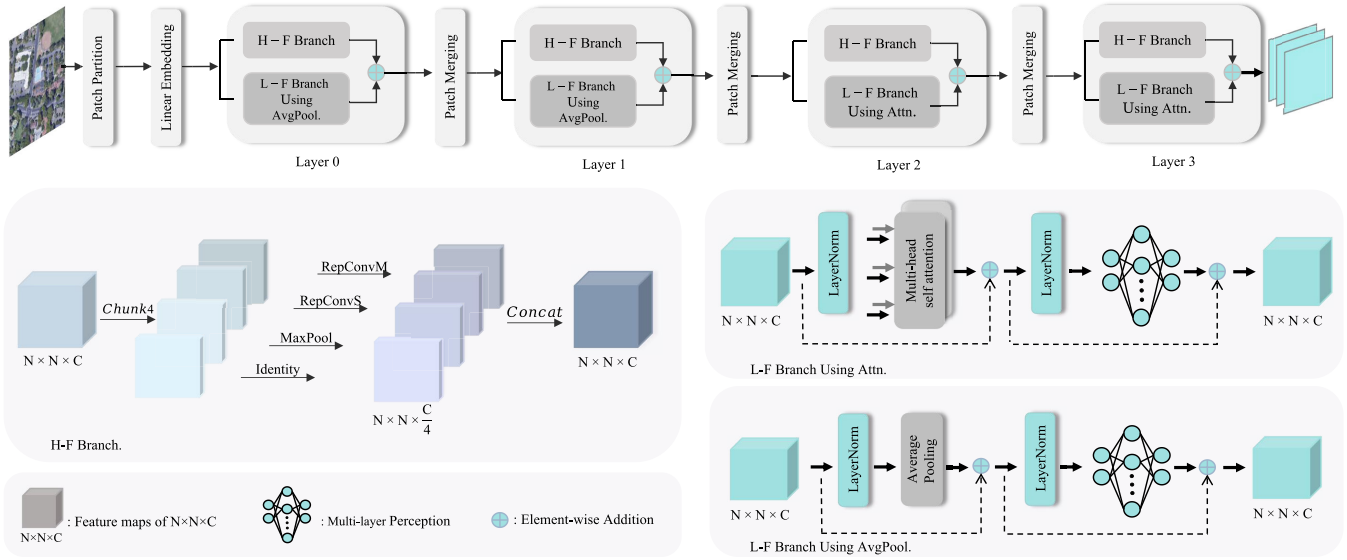


Fig. 3. Overall architecture of the proposed RingMo-Qlite. The top of the figure shows a schematic of the entire network framework, while the bottom part displays the schematics of the HFB and LFB, respectively. Among them, the LFBs of the first two layers adopt AvgPooling as the token mixer, and the LFBs of the latter two layers use the attention mechanism as the token mixer.

TABLE I
MODEL CONFIGURATIONS FOR RINGMO-QLITE

Stage	Output Size	Layer Name	RingMo-Qlite Configuration
Stage 1		Patch Embedding	Kernel: 4 × 4, Stride = 4, Embed_Dim = 96
	56 × 56 × 96	High-Low Frequency Module (HLFM)	$\begin{bmatrix}\text{num_heads} = 3\\ \text{w.s.} = 7\\ \text{H-F Branch}\\ \text{L-F Branch(Avgpool)}\end{bmatrix} \times 2$
Stage 2		Patch Merging	input_dim = 96, output_dim = 192
	28 × 28 × 192	High-Low Frequency Module (HLFM)	$\begin{bmatrix}\text{num_heads} = 6\\ \text{w.s.} = 7\\ \text{H-F Branch}\\ \text{L-F Branch(Avgpool)}\end{bmatrix} \times 2$
Stage 3		Patch Merging	input_dim = 192, output_dim = 384
	14 × 14 × 384	High-Low Frequency Module (HLFM)	$\begin{bmatrix}\text{num_heads} = 12\\ \text{w.s.} = 7\\ \text{H-F Branch}\\ \text{L-F Branch(Attn)}\end{bmatrix} \times 6$
Stage 4		Patch Merging	input_dim = 384, output_dim = 768
	7 × 7 × 768	High-Low Frequency Module (HLFM)	$\begin{bmatrix}\text{num_heads} = 24\\ \text{w.s.} = 7\\ \text{H-F Branch}\\ \text{L-F Branch(Attn)}\end{bmatrix} \times 2$
Params (M)			26.650
FLOPs (G)			3.6444

comprises four stages, each featuring a unique number of high-low frequency modules (HLFMs). The configuration follows the Swin tiny model, with the numbers (2, 2, 6, 2) representing the count of HLFMs in each stage, respectively.

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Within each HLFM, there are parallel branches for extracting low-frequency features (low-frequency branch, LFB) and high-frequency features (high-frequency branch, HFB).

The input RGB image, denoted as $I \in H \times W \times 3$, is first partitioned into nonoverlapping patches by the patch partition module, where every 4×4 set of adjacent pixels constitutes a patch

$$T = \text{PatchPartition}(I) \tag{1}$$

where $T \in H/4 \times W/4 \times 48$ denotes the vector sequence of each image patch.

These patches are then flattened along the channel dimension, and subsequently passed through a linear embedding layer, where each patch is mapped to a fixed-dimensional vector space X

$$X = \text{LinearEmb}(T) \tag{2}$$

where $X \in H/4 \times W/4 \times C$ denotes the embedding tokens with embedding dimension C.

The image representation is then obtained through the sequential processing of the four stages. Before each stage, the patch merging module reduces the image resolution and merges feature maps from adjacent levels, facilitating multiscale feature fusion.

In each HLFM, the two parallel branches, the HFB and the LFB, leverage the respective advantages of CNNs and transformers for feature extraction. LFB aims to capture long-range dependencies to obtain global features of the image, while HFB focuses on extracting high-frequency features of the image through a CNN architecture composed of four parallel structures. At the end of each HLFM, the parallel branches extract high- and low-frequency information from the image, which is then element-wise added to fuse high- and low-frequency information

$$Y_i = \text{HLFM}(X_i), \quad i = 1, 2, 3, 4 \tag{3}$$

$$\text{HLFM}(X_i) = \text{HFB}(X_i) + \text{LFB}(X_i) \tag{4}$$

where $Y_i \in H/2^{i+1} \times W/2^{i+1} \times C$ denotes the output of the four HLFMs.

B. High-Low Frequency Module

1) Low-Frequency Branch: The original Swin transformer uses window multihead self-attention (W-MSA) for local self-attention and shifted W-MSA (SW-MSA) for interwindow information exchange. However, these modules still consume significant resources. Since early layers focus on local patterns and deeper layers capture long-range dependencies, using expensive multihead attention in shallow stages may not be efficient. The model design must balance long-range dependencies with computational load and parameter count while avoiding excessive precision loss due to quantization.

In our work, the latter two stages of the LFB continue to be based on the Swin transformer architecture to capture long-range dependencies. The input features first pass through a quantized LN and then into an alternating sequence of W-MSA and SW-MSA modules $\text{MSA}_{W/\text{SW}}$, with residual connections. Subsequently, they enter another quantized LN layer and a quantized MLP, followed by another residual connection to obtain the output. In the first two stages of the LFB, as is shown in Fig. 4, a training-free 3×3 average pooling layer replaces the $\text{MSA}_{W/\text{SW}}$

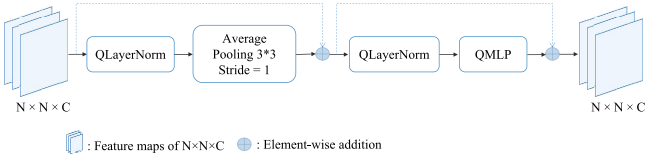


Fig. 4. Schematic of the former two stages of the LFB, where the dashed arrow represents a residual connection.

$$\begin{aligned}
\hat{\mathbf{z}}^l &= \text{MSA}_{W/\text{SW}}\left(\text{LN}\left(\mathbf{z}^{l-1}\right)\right) + \mathbf{z}^{l-1} \\
\mathbf{z}^l &= \text{MLP}\left(\text{LN}\left(\hat{\mathbf{z}}^l\right)\right) + \hat{\mathbf{z}}^l \\
\hat{\mathbf{z}}^l &= \text{AvgPool}_{3\times3}\left(\text{LN}\left(\mathbf{z}^{l-1}\right)\right) + \mathbf{z}^{l-1} \\
\mathbf{z}^l &= \text{MLP}\left(\text{LN}\left(\hat{\mathbf{z}}^l\right)\right) + \hat{\mathbf{z}}^l
\end{aligned} \tag{5}$$

where $\hat{\mathbf{z}}^l$ and $\mathbf{z}^l$ denote the output features of the $\text{MSA}_{W/\text{SW}}$ module and the MLP module for block l, respectively.

2) High-Frequency Branch: CNNs excel at extracting high-frequency details from remote sensing images, but simple architectures struggle with full quantization. Quantization reduces data precision, limiting the model's ability to capture fine details and high-frequency features. Inspired by Zhao et al. [20], our work introduces four parallel structures during the training process to enhance high-frequency extraction. During inference, these structures are simplified to reduce parameters and computational load while retaining detail capture capability.

In the high-frequency information extraction branch, we evenly divide the input features of size $F_{N\times N\times C}$ into four parts along the channel dimension, that is, F_1, F_2, F_3, F_4, each with dimensions $N \times N \times C/4$. Subsequently, a parallel architecture comprising four branches extracts different high-frequency feature information, as shown in Fig. 5. The features of these four branches are then merged by using a concatenated operation that serves as the extracted high-frequency features of the image. The setup of RepDWConvS and RepDWConvM is consistent with that of [20], comprising multiple branches during training and merging into a single branch during inference. After reparameterization, RepDWConvM and RepDWConvS achieve a reduction of more than 50% in both parameters and computational effort, which enhances the lightweight effect of the model

$$F_1, F_2, F_3, F_4 = \text{Chunk}(F) \tag{6}$$

$$\begin{aligned}
\text{HF}_1 &= \text{RepDWConvS}_{3\times3}(F_1) \\
\text{HF}_2 &= \text{RepDWConvM}_{7\times7}(F_2) \\
\text{HF}_3 &= \text{MaxPool}_{3\times3}(F_3) \\
\text{HF}_4 &= F_4
\end{aligned} \tag{7}$$

$$\text{HF} = \text{Concat}(\text{HF}_1, \text{HF}_2, \text{HF}_3, \text{HF}_4). \tag{8}$$

HF_1 and HF_2 capture high-frequency information on geospatial targets at different scales through RepDWConvS and RepDWConvM, respectively, maintaining dense computation and improving real-time inference.

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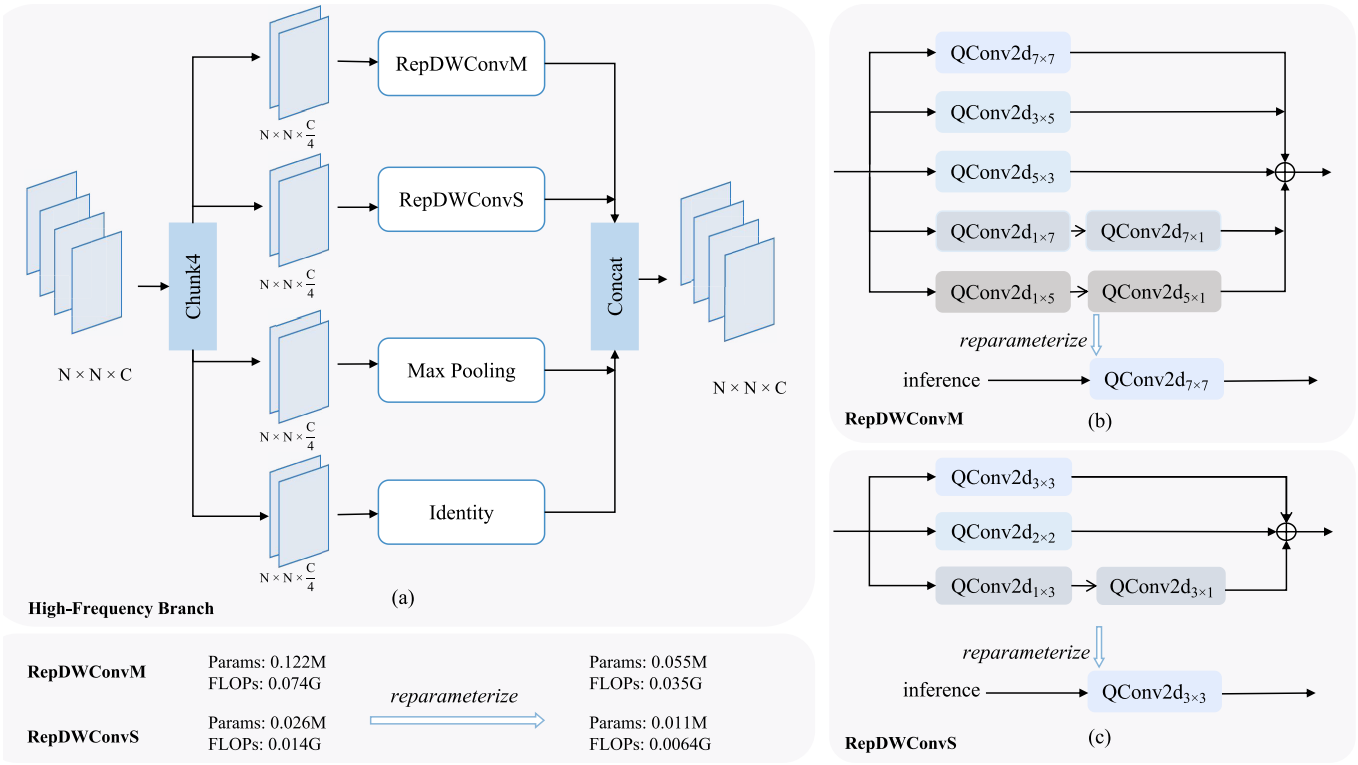


Fig. 5. Schematic of the HFB network structure: (a) presents the overall architecture of HFB, (b) presents the reparameterization schematic of RepDWConvM structure, and (c) presents the reparameterization schematic of RepDWConvS structure, with QConv2d quantized to 8-/4-bit. RepDWConvM/RepDWConvS achieves 50% reduction in Params and FLOPs after reparameterization.

C. Full-Quantization Network Low-Bit Quantization Strategy

We propose a full-quantization approach that quantizes all computational modules within the network, including both weights and activations, to perform rapid inference using only integer operations and bit shifts, eliminating the need for any floating-point computations. We adopt the FQ-ViT [18] quantization modules, including QLinear, QConv2d, QAct, QMatmul, and QLayerNorm, and utilize the Shiftmax and ShiftGELU modules of I-ViT [40] for the quantization of Softmax and GELU operations, respectively.

1) Quantization Process: We employ a PTQ strategy, utilizing a small-scale calibration dataset to perform full quantization of the model. Following the completion of model training, the distribution and statistical information of the model's weights and activations during inference are collected by an observer to determine the optimal quantization parameters. Subsequently, the model's floating-point parameters are converted into low-precision integer representations to reduce model size and enhance inference efficiency while striving to maintain model performance, thus adapting to resource-constrained on-board computing environments.

The uniform quantization scheme we adopt includes three quantization parameters: the scaling factor s, the zero-point z, and the bit-width b. The bit-width determines the range of the quantized integers (integer grid), while the scaling factor and zero-point map the floating-point numbers to this integer range. The parameters represent the minimum and maximum values of the integer range. For a signed integer range, the signed integer grid is $[-2^{b-1}, 2^{b-1}-1]$. For an unsigned integer range, the unsigned integer grid is $[0, 2^b-1]$

$$\boldsymbol{x}_{\text{int}} = \text{clamp}\left(\frac{\boldsymbol{x}}{s} + z, q_{\min}, q_{\max}\right) \tag{9}$$

$$s = \frac{\boldsymbol{x}_{\max} - \boldsymbol{x}_{\min}}{q_{\max} - q_{\min}} \tag{10}$$

$$z = \text{clamp}\left(\text{round}\left(q_{\max} - \frac{\boldsymbol{x}_{\max}}{s}\right), q_{\min}, q_{\max}\right) \tag{11}$$

$$\text{clamp}\,(x; a, c) = \begin{cases} a, & x < a \\ x, & a \le x \le c \\ c, & x > c. \end{cases} \tag{12}$$

After obtaining the quantized values, $\boldsymbol{x}_{\text{int}}$ can be dequantized to approximate the real-valued input

$$\boldsymbol{x} \approx \tilde{\boldsymbol{x}} = s\,(\boldsymbol{x}_{\text{int}} - z) \tag{13}$$

$$\hat{\boldsymbol{x}} = s \cdot \left(\text{clamp}\left(\left\lfloor \frac{\boldsymbol{x}}{s} \right\rfloor + z, q_{\min}, q_{\max}\right) - z\right). \tag{14}$$

2) Peak-Aware Quantizer for Presoftmax Quantization: Inspired by Bondarenko et al. [19], we further observe the softmax attention mechanism in transformers with a focus on inherent remote sensing scenario characteristics, and identify a key phenomenon that the inputs to the softmax layer exhibit an expanded dynamic range and contain sparse high-value peaks carrying critical semantic information, which is depicted in Figs. 6 and 7. This poses challenges for quantization, as traditional uniform quantization relies on a single global (scale and zero-point) pair and its range is dominated by the dense base region, leading to overly large quantization intervals for peaks and thus accuracy loss.

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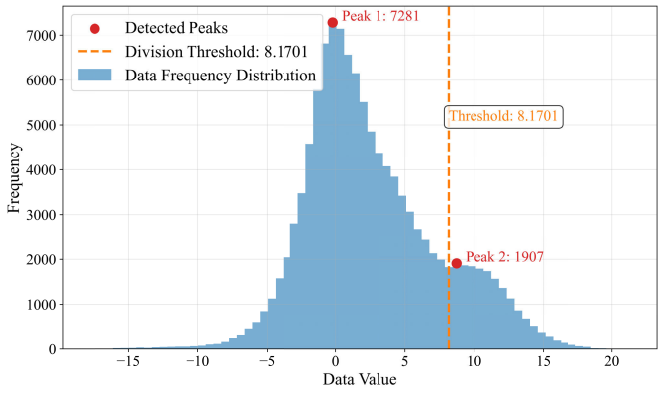


Fig. 6. Numerical distribution histogram of transformer presoftmax layer (two peaks with significant height difference annotated).

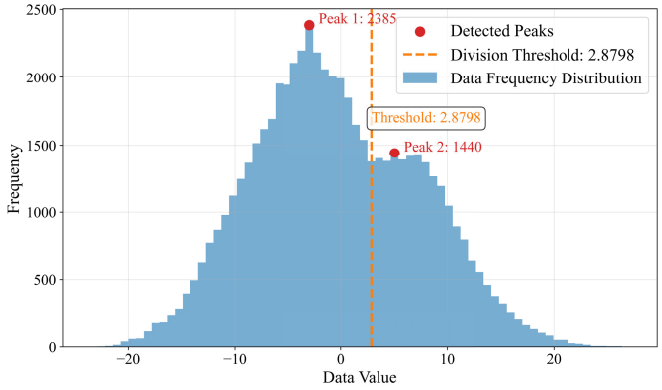


Fig. 7. Numerical distribution histogram of transformer presoftmax layer (two peaks with similar heights annotated).

To address this issue, we propose a peak-aware quantizer that adaptively partitions presoftmax data into base and peak regions, allocates differentiated quantization resources, and employs two independent sets of parameters for region-specific quantization. The implementation workflow is as follows.

First, for adaptive region division, we construct a histogram of the flattened presoftmax tensor $\boldsymbol{x}_{\text{flat}}$. Let $B = 70$ denote the total number of histogram bins; the center b_i of the ith bin and the frequency c_i of data falling into this bin are calculated as

$$b_i = \min(\boldsymbol{x}_{\text{flat}}) + \frac{i \cdot (\max(\boldsymbol{x}_{\text{flat}}) - \min(\boldsymbol{x}_{\text{flat}}))}{B} \tag{15}$$

$$c_i = \text{count}\left(\boldsymbol{x}_{\text{flat}} \in \left[b_i - \frac{\Delta}{2}, b_i + \frac{\Delta}{2}\right]\right) \tag{16}$$

where $\Delta = (\max(\boldsymbol{x}_{\text{flat}}) - \min(\boldsymbol{x}_{\text{flat}}))/B$ represents the width of each histogram bin. A bimodal detection algorithm is then applied to identify peaks in the histogram, with two constraints to ensure reliability: 1) only peaks with a frequency >10% of the maximum frequency are retained to filter low-frequency noise and 2) a minimum separation of 10 bins is enforced between adjacent peaks to avoid misclassifying local fluctuations as multiple peaks. The valley between the two strongest peaks is set as the division threshold, partitioning the original data into the base region $\boldsymbol{x}_{\text{base}}$ (values ≤ threshold) and peak region $\boldsymbol{x}_{\text{peak}}$ (values > threshold). As shown in Figs. 6 and 7, our peak-aware quantizer can find two qualifying peaks, regardless of whether the heights of these two peaks are similar or quite different.

We allocate quantization levels in a 2:1 ratio to the base and peak regions. For example, in 4-bit symmetric quantization (global range $Q_{\min} = -8, Q_{\max} = 7$), the base region uses levels $Q_{\min}^{\text{base}} = -8$ to $Q_{\max}^{\text{base}} = 2$, while the peak region uses $Q_{\min}^{\text{peak}} = 3$ to $Q_{\max}^{\text{peak}} = 7$. Region-specific (scale and zero-point) pairs are calibrated using the same observer type to ensure optimal matching with respective data ranges

$$s_{\text{base}} = \frac{\max(\boldsymbol{x}_{\text{base}}) - \min(\boldsymbol{x}_{\text{base}})}{Q_{\max}^{\text{base}} - Q_{\min}^{\text{base}}} \tag{17}$$

$$z_{\text{base}} = Q_{\min}^{\text{base}} - \frac{\min(\boldsymbol{x}_{\text{base}})}{s_{\text{base}}} \tag{18}$$

$$s_{\text{peak}} = \frac{\max(\boldsymbol{x}_{\text{peak}}) - \min(\boldsymbol{x}_{\text{peak}})}{Q_{\max}^{\text{peak}} - Q_{\min}^{\text{peak}}} \tag{19}$$

$$z_{\text{peak}} = Q_{\min}^{\text{peak}} - \frac{\min(\boldsymbol{x}_{\text{peak}})}{s_{\text{peak}}}. \tag{20}$$

Quantization is performed separately for each region, with an offset $\Delta Q = Q_{\max}^{\text{base}} - Q_{\max}^{\text{peak}'}$, where $Q_{\max}^{\text{peak}'}$ is the original upper bound of the peak region before offset, to avoid overlapping quantized values

$$\boldsymbol{q}_{\text{base}} = \text{clamp}\left(\text{round}\left(\frac{\boldsymbol{x}_{\text{base}}}{s_{\text{base}}} + z_{\text{base}}\right), Q_{\min}^{\text{base}}, Q_{\max}^{\text{base}}\right) \tag{21}$$

$$\boldsymbol{q}_{\text{peak}} = \text{clamp}\left(\text{round}\left(\frac{\boldsymbol{x}_{\text{peak}}}{s_{\text{peak}}} + z_{\text{peak}}\right) + \Delta Q, Q_{\min}^{\text{peak}}, Q_{\max}^{\text{peak}}\right). \tag{22}$$

Dequantization reverses the above process, first canceling the offset for peak-region quantized values before recovering the original data distribution

$$\hat{\boldsymbol{x}}_{\text{base}} = (\boldsymbol{q}_{\text{base}} - z_{\text{base}}) \cdot s_{\text{base}} \tag{23}$$

$$\hat{\boldsymbol{x}}_{\text{peak}} = (\boldsymbol{q}_{\text{peak}} - \Delta Q - z_{\text{peak}}) \cdot s_{\text{peak}}. \tag{24}$$

By decoupling the quantization of base and peak regions, this design ensures that critical high-value features in presoftmax inputs are preserved under low-bit constraints, effectively mitigating the accuracy degradation of quantized transformer models for remote sensing tasks.

3) Outlier-Robust Observer for Quantization Parameter Calibration: In the PTQ workflow, a small-scale calibration dataset is employed to calibrate quantization parameters. The Observer collects statistical information of the model's weights and activations during inference, which serves as the basis for determining optimal quantization parameters (scale and zero-point). However, remote sensing images frequently contain anomalous brightness values (e.g., strongly reflective ground objects and sensor noise points), which distort the statistical characteristics of weights/activations and thus lead to suboptimal quantization parameters. Existing methods, such as the approach proposed in [18], utilize the L2 norm to calculate the error between original and quantized data. This metric is highly sensitive to outliers, resulting in poor robustness in remote sensing scenarios. To address this issue, we design a novel Observer mechanism to adaptively optimize quantization parameters for each network layer. Drawing inspiration from

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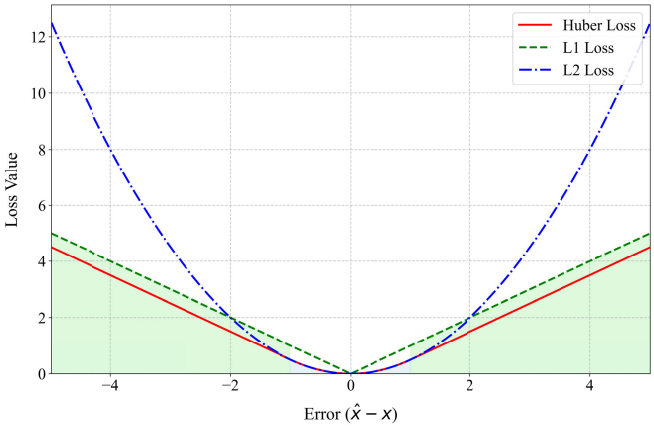


Fig. 8. Comparison of Huber loss with L1 and L2 loss.

the iterative refinement strategy in [41], the proposed Observer adjusts the range of maximum and minimum values for each layer independently. Specifically, Huber loss is adopted to replace the traditional L2 norm as the optimization objective, which mitigates outlier interference in quantization parameter calibration

$$\begin{aligned} \boldsymbol{x}'_{\max} &= \gamma \cdot \boldsymbol{x}_{\max} \\ \boldsymbol{x}'_{\min} &= \gamma \cdot \boldsymbol{x}_{\min} \\ s' &= \gamma \cdot s \\ z' &= \text{clamp}\left(\text{round}\left(q_{\max} - \frac{x'_{\max}}{s'}\right), q_{\min}, q_{\max}\right) \end{aligned} \tag{25}$$

$$\hat{x}' = s' \cdot \left(\text{clamp}\left(\left\lfloor \frac{\boldsymbol{x}}{s'} \right\rfloor + z', q_{\min}, q_{\max}\right) - z'\right). \tag{26}$$

We use the Huber loss function to combine the L1 and L2 norms, applying L2 for small errors and L1 for larger errors, as depicted in Fig. 8. This helps maintain smoothness for minor quantization errors and reduces the impact of outliers in significant errors. Using Huber loss effectively reduces performance degradation from low-bit quantization, enhancing robustness and accuracy, especially for remote sensing images with varying brightness and contrast. By minimizing the Huber loss, the proposed observer retains valid statistical information of features while reducing the interference of outliers in remote sensing images on quantization parameter calibration. This ensures that each layer obtains more robust scale and zero-point parameters, laying a foundation for improving the accuracy of low-bit quantized remote sensing models

$$\mathcal{L}_{\text{Huber}}(\hat{x}, x) = \begin{cases} \frac{1}{2}(\hat{x} - x)^2, & \text{if } |\hat{x} - x| \le \delta \\ \delta\left(|\hat{x} - x| - \frac{1}{2}\delta\right), & \text{otherwise} \end{cases} \tag{27}$$

where $\hat{x}$ denotes the dequantized value, x represents the original value, and δ is the switching threshold of the loss function (set to 1.0 in experiments).

IV. Experiments

A. Dataset Introduction

We comprehensively evaluate the image interpretation capabilities of our proposed fully quantized remote sensing foundation model on three classic remote sensing scene classification datasets.

1) UCM [42]: The UCM dataset comprises 2100 aerial scene images categorized into 21 land-use scene classes. Each class contains 100 aerial images, with a resolution of 256 × 256 pixels per image and a spatial resolution of 0.3 m/pixel in the red–green–blue space. Extracted from aerial orthoimages in the National Map of the United States Geological Survey (USGS), these images originate from various cities across the United States. The dataset includes highly overlapping land-use categories, such as dense, medium, and sparse residential areas, which primarily differ in structural density, making the dataset more diverse and challenging. It is widely used for remote sensing image scene classification and retrieval tasks.

2) AID [43]: The AID dataset contains images sourced from Google Earth, encompassing 30 different types of aerial scenes with a total of 10 000 images. Each category ranges from 220 to 420 images, with a resolution of 600 × 600 pixels. The sample images in the AID dataset are selected from different countries and regions worldwide, mainly in China, the United States, the U.K., France, Italy, Japan, and Germany. Captured under various seasons and imaging conditions, these images increase intraclass diversity. In addition, the images are captured by different remote sensing imaging sensors, adding to the challenge of remote sensing scene classification tasks due to their multisource nature.

3) NWPU-RESISC45 [44]: The NWPU-RESISC45 dataset includes 31 500 remote sensing images divided into 45 scene categories. Each category contains 700 images of 256 × 256 pixels with spatial resolutions ranging from approximately 30–0.2 m/pixel. Extracted by experts in the field of remote sensing image interpretation from Google Earth, these 31 500 images cover more than 100 countries and regions around the world. The dataset is large-scale, rich in variation, and characterized by high intraclass diversity and interclass similarity, introducing finer-grained scene categories with high semantic overlap, and presenting new challenges to remote sensing image scene classification tasks.

B. Detailed Experimental Setup

All experiments are conducted using Docker images and the PyTorch framework on an NVIDIA GeForce RTX 4090 GPU. The batch size is set to 128, with training continuing for an average of 400 epochs. Model optimization is performed using the Adam optimizer with hyperparameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. Following the pretraining method of RingMo-lite [17], weights are loaded from 150 000 visible images processed through masked image modeling (MIM). The calibration dataset for PTQ is composed of approximately ten samples randomly selected from the training set of each benchmark dataset, following the distribution of the training set to ensure representative data characteristics. For fair comparison, datasets are divided as in RingMo-lite, with training sample ratios of TR = 10% and TR = 20% for NWPU-RESISC, TR = 20% and TR = 50% for AID, and TR = 80% for UCM. Each experiment is repeated five times to ensure robustness. Overall accuracy (OA) is used as the performance metric for remote sensing image scene classification tasks.

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TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS ACROSS VARIOUS DATASETS AND TRAINING RATIOS (TRS)

Method	AID TR=20%	AID TR=50%	NWPU-RESISC45 TR=10%	NWPU-RESISC45 TR=20%	UCM TR=80%
MobileNet-v2 [45]	73.82	85.66	68.40	79.13	92.86
MobileNet-v3s [46]	75.15	82.54	63.39	74.92	91.90
MobileNet-v3l [46]	74.12	86.16	67.21	78.45	92.38
ShuffleNet-v2-x1 [47]	72.29	85.10	64.57	75.81	91.19
ShuffleNet-v2-x2 [47]	76.91	87.78	69.70	80.35	92.86
EfficientNet-b0 [48]	75.06	85.72	69.23	80.50	92.62
EfficientNet-b7 [48]	85.01	92.32	85.27	86.06	95.95
MobileViT-xs [23]	78.94	88.94	71.44	80.66	94.76
MobileViT-s [23]	79.29	88.82	71.89	78.98	95.48
ResNet-50 [49]	77.34	88.06	68.51	80.75	93.57
Swin Tiny(MIM) [22]	89.04	95.18	86.67	91.62	95.71
RingMo-lite [17]	93.86	96.54	89.85	93.25	99.05
RingMo-Qlite	**94.41**(↑0.6%)	**97.06**(↑0.5%)	**90.51**(↑0.6%)	**93.94**(↑0.7%)	**99.29**(↑0.2%)

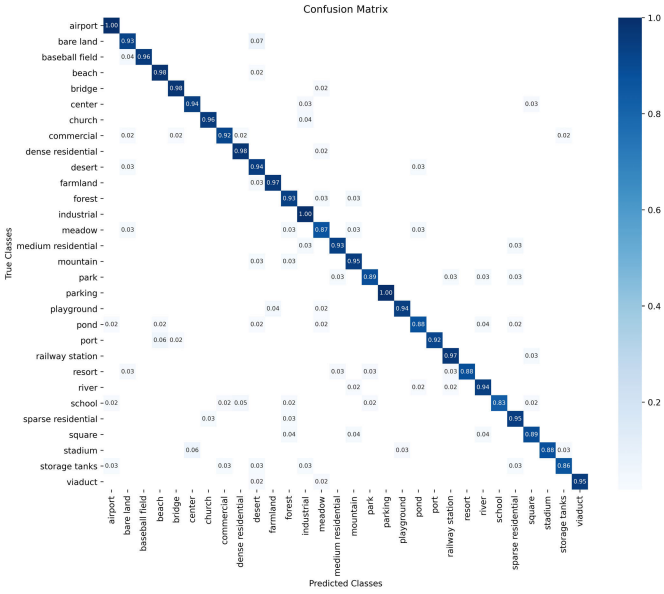


Fig. 9. Classification confusion matrix for RingMo-Qlite on 20% AID training data.

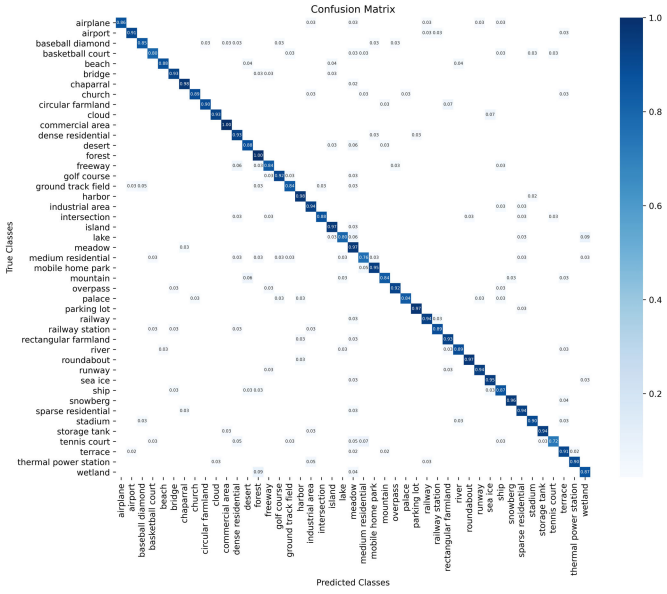


Fig. 10. Classification confusion matrix for RingMo-Qlite on 10% NWPU-RESISC45 training data.

C. Experimental Result Analysis

1) Network Performance Before Full Quantization: Table II presents the results of our proposed RingMo-Qlite on various remote sensing scene classification datasets, compared with other lightweight methods. RingMo-lite, which serves as the baseline for our method, is included in this comparison. As demonstrated in Table II, our method achieves the best classification outcomes across all datasets and under various training sample ratios. Specifically, our method shows a significant improvement over the baseline, with enhancements of 0.55% in the challenging AID (TR = 20%) and 0.66% in the NWPU (TR = 10%) classification tasks, thereby validating the superiority of our approach. The confusion matrices in Figs. 9 and 10 provide a detailed visualization of the performance of our classification model and show obvious misclassification between highly feature-similar scenarios such

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TABLE III
COMPARISON OF DIFFERENT METHODS REGARDING PARAMETERS, FLOPS, AND OA ON UCM DATASET

Method	Para. (M)	FLOPs (G)	OA (%)
ResNet-50 [49]	**23.551**	4.132	93.57
Swin Tiny [22]	28.288	4.494	95.71
RingMo-lite [17]	28.294	4.494	99.05
RingMo [50]	87.768	15.438	99.06
RingMo-Qlite	26.650 (↑13.2%)	**3.644** (↓11.8%)	**99.29** (↑0.2%)

as tennis court and basketball court, meadow and bare land, as well as spatially structure-related scenarios such as school and commercial. This is mainly due to the challenge of inherent feature similarity in remote sensing scenes and the further optimization space of the model in representing fine-grained features and scene functional attributes.

Table III compares our method with other existing lightweight methods in terms of parameter count and inference-time computational load, with all comparisons conducted prior to full quantization. Computational load is measured by the number of FLOPs, and both parameter count and FLOPs are quantified using the THOP library. The reported OA corresponds to the classification performance on the UCM dataset with a training sample ratio of TR = 80%. It is evident that our method outperforms other lightweight counterparts through a more advanced lightweight design: compared to existing lightweight models, it not only reduces the model's FLOPs by 18.9% (achieving lower computational load) and cuts parameter count by 5.8%, but also further boosts the OA on the UCM dataset to 99.29%. This clearly demonstrates that even before full quantization, our network already delivers smaller FLOP volumes and higher accuracy than current lightweight methods.

2) Full-Quantization Network Performance: Table IV shows the performance of our quantization method across different datasets under various quantization settings and TRs. We analyze the performance of our proposed RingMo-Qlite and existing PTQ methods under different quantization settings on the AID dataset [43], NWPU-RESISC45 dataset [44], and UCM dataset [42], and the results are shown in Table V. It is worth noting that the integer-only implementation constraint and bit configurations (for weights and activations) are explicitly marked in the table as key conditional factors. Compared with other methods, the integer-only design implies a more stringent implementation restriction, requiring model inference to be performed entirely through integer arithmetic. In contrast, other existing methods avoid integer quantization of nonlinear layers and only quantize the weights and activations in the network. We focus on the performance under low-bit quantization, including W8/A8 (8 bits for weights and 8 bits for activations), W4/A8 (4 bits for weights and 8 bits for activations), and W4/A4 (4 bits for weights and 4 bits for activations) settings to highlight the performance advantages of RingMo-Qlite under such strict constraints.

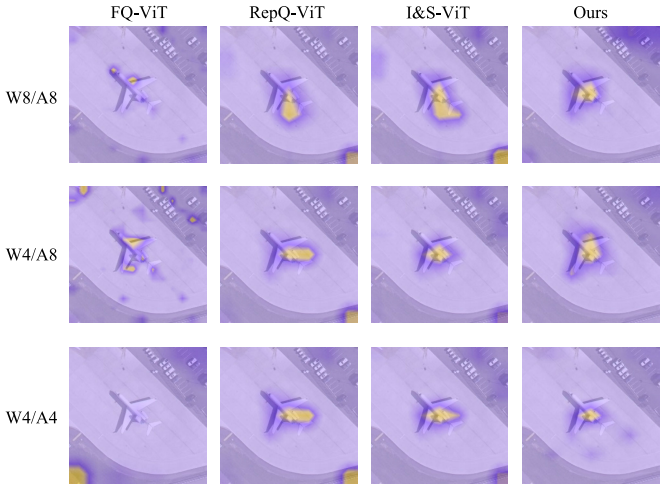


Fig. 11. Attention heatmap comparison of different quantization schemes and methods (row-wise: W8/A8, W4/A8, and W4/A4 quantization settings; column-wise: FQ-ViT, RepQ-ViT, I&S-ViT, and our method).

In the two scenarios of W8/A8 quantization and W4/A8 quantization, our method can outperform other existing methods, demonstrating strong quantization adaptability. In the challenging scenario of W4/A4 quantization, FQ-ViT [18] is almost inoperable, achieving an accuracy of only 0.10% on multiple datasets; RepQ-ViT [38] and I&S-ViT [39] perform relatively well. They only quantize the weights and activations in the network, which can avoid the accuracy loss caused by directly using integer inference in nonlinear layers, but there is still an obvious performance loss. In contrast, despite being restricted by the integer-only implementation, RingMo-Qlite still shows strong quantization adaptability. Under the W4/A4 quantization setting, its accuracy on the AID dataset (training set ratio 50%), NWPU-RESISC45 dataset (training set ratio 20%), and UCM dataset (training set ratio 80%) reaches 88.91%, 86.75%, and 92.36%, respectively, which can approach the accuracy of I&S-ViT [39]. These results show that RingMo-Qlite not only meets the more stringent integer-only implementation conditions but also can approach or even surpass the performance of existing noninteger-only methods under this constraint, verifying the effectiveness and superiority of its design. We visualize the attention maps of various methods under different quantization settings, as shown in Fig. 11. It can be observed that when using 8-bit quantization, all methods can basically focus on the target region. As the quantization bit-width decreases, other methods exhibit significantly reduced attention focus and increased background interference, while our method can consistently maintain stable focus on the target and preserve good feature representation performance.

D. Ablation Study

We have conducted a series of ablation experiments on the modules within our proposed method. Our comprehensive analysis has led to the following conclusions.

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TABLE IV
PERFORMANCE OF OUR QUANTIZATION METHOD ACROSS DIFFERENT DATASETS AND TRS UNDER VARIOUS QUANTIZATION SETTINGS

Bit. (W/A)	AID TR=20%	AID TR=50%	NWPU-RESISC45 TR=10%	NWPU-RESISC45 TR=20%	UCM TR=80%
32/32	94.41	97.06	90.51	93.94	99.29
8/8	92.63	95.46	89.14	91.96	98.85
4/8	91.81	95.11	88.58	91.02	98.09
4/4	86.59	88.91	84.33	86.75	92.36

TABLE V
PERFORMANCE COMPARISON OF DIFFERENT QUANTIZATION METHODS ON REMOTE SENSING DATASETS WITH DIFFERENT BIT-WIDTHS, INTEGER-ONLY INFERENCE SUPPORT, AND TRS

Method	Int.-only	Bit. (W/A)	AID TR=50%	NWPU-RESISC45 TR=20%	UCM TR=80%
Full-Precision	-	32/32	97.06	93.94	99.29
FQ-ViT [18]	✗	8/8	93.81	88.17	97.89
RepQ-ViT [38]	✗	8/8	94.96	90.24	98.28
I&S-ViT [39]	✗	8/8	95.12	90.77	98.76
RingMo-Qlite (Ours)	✓	8/8	**95.46**(↑0.3%)	**91.96**(↑1.3%)	**98.85**(↑0.1%)
FQ-ViT [18]	✗	4/8	90.19	84.29	92.33
RepQ-ViT [38]	✗	4/8	92.35	88.43	95.10
I&S-ViT [39]	✗	4/8	94.86	**91.26**	97.48
RingMo-Qlite (Ours)	✓	4/8	**95.11**(↑0.3%)	91.02(↓0.3%)	**98.09**(↑0.6%)
FQ-ViT [18]	✗	4/4	0.10	0.10	0.10
RepQ-ViT [38]	✗	4/4	88.67	86.33	92.81
I&S-ViT [39]	✗	4/4	**89.32**	**86.98**	**92.86**
RingMo-Qlite (Ours)	✓	4/4	88.91(↓0.5%)	86.75(↓0.3%)	92.36(↓0.5%)

1) LFB and HFB: As can be seen from Fig. 12, after quantization, the overall distribution trend of features remains highly consistent with that of the unquantized model, and the core detail-capturing capability is barely impaired.

We present the comparative experimental results of the number of branches in the HFB, as shown in Table VI. We tested three other variants: Variant 1 (2 branches, only small/medium convolutional branches), Variant 2 (3 branches, adding MaxPool to the 2-branch setup), and Variant 3 (5 branches, expanding to large-scale convolution). Our 4-branch variant achieves the best balance between performance and efficiency on the UCM dataset.

The HFB enhances the network's ability to capture fine-detail features by integrating parallel CNN feature extraction modules. In addition, reparameterization techniques boost the network's robustness against quantization noise. As demonstrated in Table VII, adding the HFB to RingMo-lite yields superior classification performance across various quantization settings. Notably, under INT8 quantization, the OA on the UCM dataset increased by 4.082%.

For low-frequency features, the early stage outputs almost completely overlap with the unquantized version, preserving the overall structural information, with only minor variations observed in later stages. This confirms that quantization hardly compromises the model's extraction of high- and low-frequency features, achieving a good balance between feature retention and model compression.

The LFB mitigates performance degradation caused by full-network quantization without sacrificing classification accuracy. It realizes this by replacing the multihead self-attention structure in the early feature extraction stage with pooling layers. As shown in Table VII, under different quantization bit settings (W8/A8 and W4/A8), the classification performance outperforms the original method. Meanwhile, Figs. 13 (low-frequency feature maps) and 14 (high-frequency feature maps) visualize that the quantized

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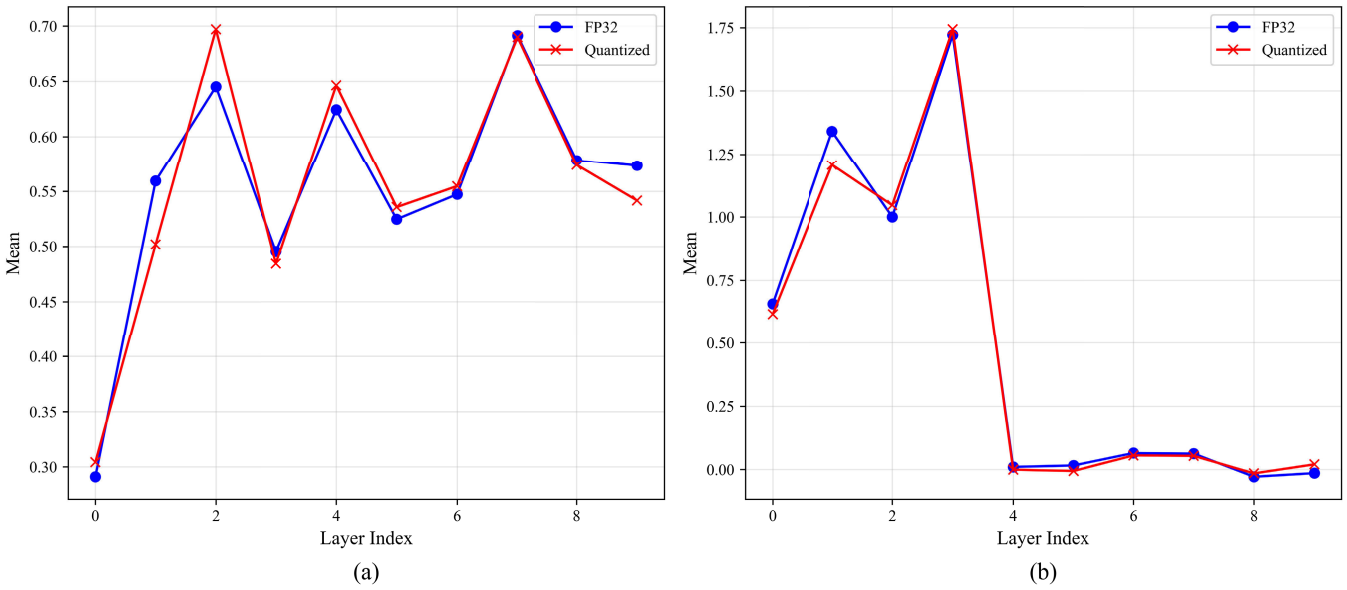


Fig. 12. Cross-layer mean comparison of features: (a) high-frequency and (b) low-frequency between prequantization and INT8 quantization.

TABLE VI
EFFECT OF THE NUMBER OF HFB BRANCHES. WE EVALUATE THE PERFORMANCE AND EFFICIENCY OF QUANTIZED RINGMO-QLITE ON THE UCM DATASET

Number of Branches	Configuration	Float32	W8/A8
2	RepDWConvS+RepDWConvM	98.63	97.51
3	RepDWConvS+RepDWConvM+MaxPool	99.01	98.24
4 (Ours)	**RepDWConvS+RepDWConvM+MaxPool+Identity**	99.29	**98.85**
5	RepDWConvS+RepDWConvM+MaxPool+Identity+RepDWConvL	**99.31**	98.10

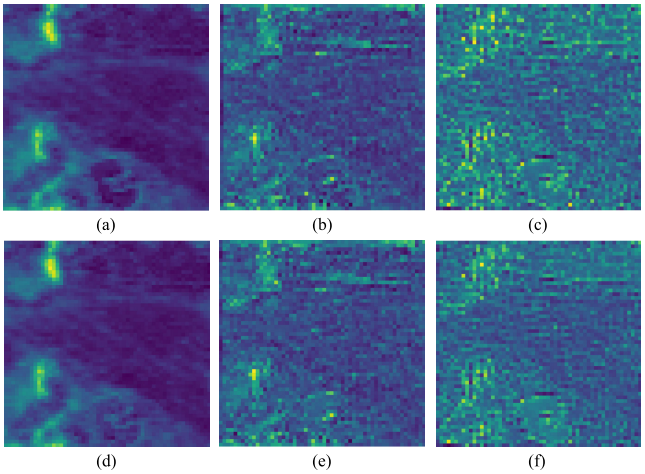


Fig. 13. Visualization of LFB features across three channels. (a)–(c) Show the prequantization features (top row), and (d)–(f) show the corresponding INT8-quantized features (bottom row).

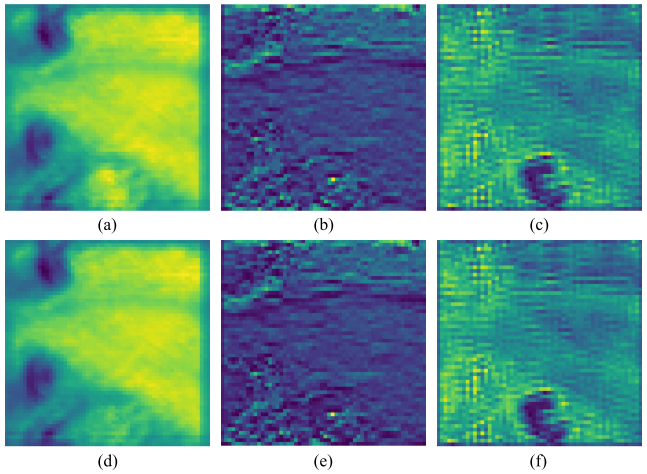


Fig. 14. Visualization of HFB features across three channels. (a)–(c) Show the prequantization features (top row), and (d)–(f) show the corresponding INT8-quantized features (bottom row).

model's outputs are highly similar to the FP32 model's. Low-frequency maps retain consistent overall structural patterns, while high-frequency maps preserve fine-grained details, further validating that quantization has a negligible impact on both types of feature extraction.

2) Outlier-Robust Observer: To investigate the impact of the outlier-robust observer on quantized RingMo-Qlite, we

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TABLE VII

COMPARISON OF THE OA OF DIFFERENT METHODS ON THE UCM DATASET

Method	Float32	W8/A8	W4/A8
Swin Tiny [22]	95.71	90.524	90.112
Swin Tiny+LFB [22]	94.39	90.866	90.295
RingMo-lite [17]	99.05	93.143	94.333
RingMo-lite+LFB [17]	97.07	93.048	94.495
RingMo-lite+HFB [17]	99.21	97.225	97.143

TABLE VIII

EFFECT OF THE OUTLIER-ROBUST OBSERVER ON DIFFERENT DATASETS. WE EVALUATE THE PERFORMANCE OF QUANTIZED RINGMO-QLITE UNDER DIFFERENT QUANTIZATION SETTINGS

Bit. (W/A)	Outlier-Robust Observer	AID TR=50%	UCM TR=80%
8/8	✓	95.46	99.29
8/8	✗	94.71	98.52
4/8	✓	95.11	98.09
4/8	✗	93.89	96.94
4/4	✓	88.91	92.36
4/4	✗	83.71	85.14

TABLE IX

EFFECT OF OUTLIER-ROBUST OBSERVER PARAMETER δ ON OA PERFORMANCE. WE EVALUATE THE OA OF QUANTIZED RINGMO-QLITE UNDER DIFFERENT δ VALUES AND QUANTIZATION SETTINGS

δ	W4/A4	W4/A8	W8/A8
0.2	88.42	96.73	97.93
0.4	89.76	97.21	97.26
0.6	90.83	97.65	96.51
0.8	91.65	97.88	97.72
1.0	**92.36**	**98.09**	**98.85**
1.2	91.98	97.95	98.79
1.4	91.24	95.72	97.12
1.6	88.57	94.46	95.48
1.8	85.81	91.19	95.32
2.0	86.03	90.92	93.15

conducted ablation experiments under different quantization bit. "Bit. (W/A)" indicates that the bit-width of the weights and activations are W and A bits, respectively. As shown in Table VIII, the outlier-robust observer consistently improved model performance on both AID and UCM datasets across different quantization configurations. Across various quantization bit settings (W8/A8, W4/A8, and W4/A4), enabling our outlier-robust observer resulted in higher accuracies on both datasets compared to disabling it. Notably, the performance gap between enabling and disabling the observer became more significant as the quantization bit-width decreased, demonstrating the observer's effectiveness in improving the robustness of quantized models. We also analyzed the relationship between the parameter δ (associated with the observer) and the OA, with results shown in Table IX. As δ varied, the OA performance of the model under different quantization configurations (W4/A4, W4/A8, and W8/A8) first increased and then decreased. Specifically, when $\delta = 1.0$, the model achieved the highest OA in all quantization settings, which suggests that this δ value optimizes the balance between the observer's outlier suppression and feature preservation capabilities.

TABLE X

EFFECT OF THE PEAK-AWARE QUANTIZER ON DIFFERENT DATASETS. WE EVALUATE THE PERFORMANCE OF QUANTIZED RINGMO-QLITE UNDER DIFFERENT QUANTIZATION SETTINGS

Bit. (W/A)	Peak-Aware Quantizer	AID TR=50%	UCM TR=80%
8/8	✓	95.46	99.29
8/8	✗	95.30	98.78
4/8	✓	95.11	98.09
4/8	✗	94.76	97.35
4/4	✓	88.91	92.36
4/4	✗	86.55	90.47

TABLE XI

EFFECT OF DIFFERENT RATIOS ON OA PERFORMANCE. WE EVALUATE THE PERFORMANCE OF QUANTIZED RINGMO-QLITE UNDER DIFFERENT QUANTIZATION SETTINGS

Ratio	W4/A4	W4/A8	W8/A8
1:3	24.29	39.05	56.67
1:2	52.32	61.43	70.71
1:1	79.08	89.29	88.33
2:1(ours)	**92.36**	**98.09**	**98.85**
3:1	89.28	95.92	92.38

3) Peak-Aware Quantizer: To explore how the Peak-Aware Quantizer influences the quantized RingMo-Qlite model, we conducted experiments under different quantization settings, with results shown in Table X. We also present the experimental results regarding the quantization level ratio between the base region and the peak region, as shown in Table XI. It can be concluded that the 2:1 ratio achieves the highest OA in all configurations, representing the optimal balance between peak and base region quantization precision. When the quantization bit was set to W8/A8, W4/A8, or W4/A4, the model consistently achieved higher accuracies with the peak-aware quantizer in use. The most significant improvements were observed under more aggressive quantization settings,

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such as the W4/A4 bit-width, where the accuracy gains were substantially larger than those under the relatively mild W8/A8 bit quantization. This pattern indicates that the peak-aware quantizer is particularly effective in preserving critical semantic information carried by sparse high-value features in remote sensing data, an aspect where traditional global quantization struggles under low-bit constraints. Thus, its enhancement becomes more pronounced as the quantization pressure increases, making it crucial to maintain performance in resource-constrained remote sensing scenarios.

E. Discussion

Through a series of experiments and comparative validations, we demonstrate that the proposed RingMo-Qlite network and its novel full-quantization method can adapt to remote sensing scenarios under various quantization settings, achieving competitive performance. Moreover, the network maintains high accuracy even after low-precision quantization, laying the foundation for remote sensing image interpretation tasks in on-orbit scenarios. This achievement is attributed to the design of the HLFB dual-branch structure and the full-quantization method that accounts for the characteristics of numerical distribution in remote sensing scenarios. The fully quantized RingMo-Qlite network is expected to provide valuable references for the development of fully quantized models in future remote sensing tasks.

However, RingMo-Qlite still faces several challenges.

1) Model Performance Under Extreme Low-Bit Quantization Demands Exploration: While the network achieves near-lossless accuracy under INT8 quantization and maintains usable performance under INT4 quantization, its behavior at bit-widths lower than 4 bits, such as 3 bits or 2 bits, remains unexamined. Such extremely low-bit settings are highly desirable for ultra resource-constrained on-orbit platforms. In future work, we will focus on structural optimizations such as adaptive bit-width allocation across network branches and refine the quantization strategy to enhance feature preservation under extreme discretization, with the aim of pushing the limits of low-bit performance.

2) Compatibility With Multisensor Remote Sensing Data Is Limited: Current RingMo-Qlite is optimized for RGB remote sensing images, while on-orbit platforms often acquire multimodal data such as hyperspectral SAR and thermal infrared images. These data types have distinct numerical characteristics, for example, hyperspectral data has hundreds of channels with narrow spectral bands, and SAR data has inherent speckle noise, which the existing quantization observer and branch structure cannot fully accommodate. Future work will extend the network to a multimodal backbone design modality-specific quantization parameters, and optimize the branch feature fusion strategy to adapt to diverse sensor data.

V. Conclusion

This article proposes RingMo-Qlite, a fully quantized lightweight CNN-transformer hybrid framework that better addresses the challenges of remote sensing image interpretation on remote sensing satellite platforms. The framework aims to reduce the computational complexity of the model without sacrificing processing accuracy, thereby enabling efficient and high-precision intelligent processing of on-board remote sensing images. We employ lightweight design strategies for the transformer architecture and reparameterization techniques for the CNN architecture to enhance feature extraction capabilities during network training while achieving accurate and fast inference. Compared to models of similar size, the proposed method achieves state-of-the-art performance in scene classification tasks. To adapt to the characteristics of remote sensing images, we design a novel quantization strategy that incorporates a peak-aware quantizer and an outlier-robust observer, aiming to minimize precision loss to the greatest extent. Our fully quantized network achieves desirable inference performance under the INT8 quantization setting (less than 1% accuracy loss) and comparable results under the INT4 quantization setting.

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